

IP Subnet Calculator

Submitted By

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1. Motivation

Network administration and infrastructure design are fundamental skills in modern computing. The motivation behind SubnetLab stems from this gap: a need for a free, open, and pedagogically valuable tool that handles every aspect of IPv4 subnetting, from the most basic host range calculations to advanced VLSM allocation and supernetting, while also showing the user exactly how each result is derived. This tool is designed not just to compute, but to educate.

2. Overview

2.1 Significance of the Project

IP subnetting is a cornerstone of network engineering. Errors in subnet planning lead to address conflicts, and security vulnerabilities. SubnetLab addresses this by providing a unified, browser accessible tool built entirely with manual binary logic, offering both correctness and transparency. Its academic value lies in the step-by-step breakdown feature, which mirrors classroom problem-solving techniques.

2.2 Description of the Project

SubnetLab is a web-based IPv4 subnet calculator built using Python Flask on the backend and plain HTML/CSS/JavaScript on the frontend. It accepts standard CIDR notation as input and provides a comprehensive suite of network tools across seven modules: Basic Subnet Calculation, Subnet Splitting, VLSM Allocation, Batch Processing, Overlap Detection, Supernetting/Aggregation, and IP Lookup. All binary and subnet logic is implemented manually without relying on high-level IP libraries.

2.3 Background of the Project

The project draws on foundational concepts from RFC 950 (Subnetting), RFC 1519 (CIDR), and RFC 4632 (Classless Inter-Domain Routing). Python's ipaddress module is used for network object modeling, while all subnet arithmetic (AND, OR, NOT masking) is performed at the bit level.

2.4 Project Category

This project falls under the Product-Based category. It is a standalone web application developed as an academic tool with practical professional utility for network engineers, students, and system administrators.

3. Features / Scope / Modules

- **1. Basic Subnet Calculator**
  - Accepts any CIDR input and computes: Network Address, Broadcast Address, Subnet Mask, Wildcard Mask, First/Last Usable Host, Total and Usable Host Count, IP Class (A/B/C/D/E), Private/Public classification, and Binary Representation of the IP, Mask, and Network..
- **2. Subnet Splitter**

    - Takes a parent network and splits it into N equal subnets. Automatically computes the required new prefix length.
- **3. VLSM (Variable Length Subnet Masking) Allocator**
    - Accepts a pool CIDR and a list of department names with host requirements. Allocates subnets in descending host-count order (largest first), assigning the tightest fitting block to each requirement.
- **4. Batch Calculator**
    - Accepts multiple CIDRs (one per line) and produces a consolidated results table with all subnet fields for each entry. Supports CSV export for use in spreadsheets and documentation.
- **5. Overlap / Conflict Detector**
    - Checks a list of CIDRs for any pairwise overlapping address ranges. Reports conflicting pairs individually with clear color-coded status, helping identify misconfigured networks before deployment.
- **6. Supernetting / Route Aggregation**
    - Takes multiple network CIDRs and collapses them into the minimum set of summary routes using address collapsing. Useful for BGP route summarization and reducing routing table size.
- **7. IP Subnet Lookup**
    - Given any IP address and a subnet CIDR, determines whether the IP falls within the subnet range. Returns a clear yes/no status with color feedback.

4. Project Planning

Week	Task	Deliverable
Week 1	Backend logic: subnet math, VLSM, overlap, supernet API endpoints	app.py with 7 REST endpoints
Week 2	Frontend UI: sidebar layout, tab panels, CSS design system, JS logic	Full working UI in browser
Week 3	Integration testing, CSV export, step-by-step rendering, final polish	Complete deployable project

5. Project Feasibility

- **Technical Feasibility**
  Fully achievable with existing free technologies. Python Flask is a mature, well-documented framework. All subnet logic is implementable using Python's standard library.
- **Economic Feasibility**
  Development cost is zero, all tools used (Python, Flask, HTML/CSS/JS) are open-source. Deployment can be done on any machine..
- **Schedule Feasibility**

The project was designed to be completed within a 3-week sprint, well within the semester timeline. The modular architecture allowed parallel development and independent testing of each module.

6. Hardware and Software Requirements

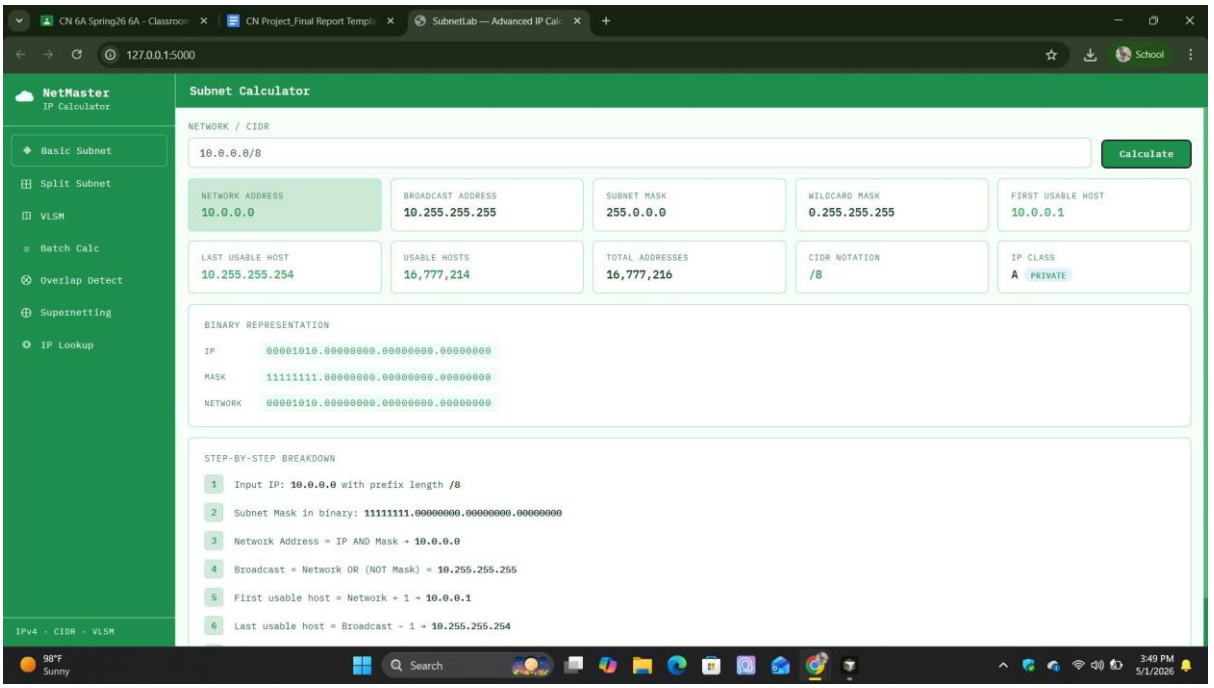
Software Requirements

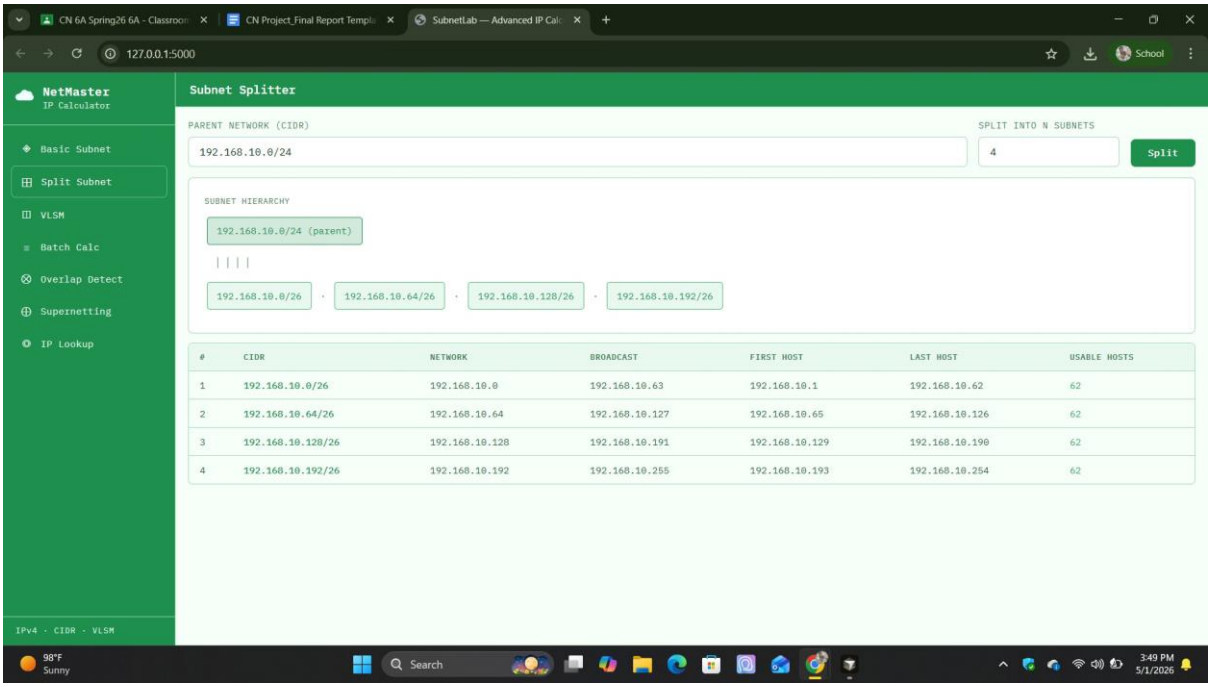
Python 3.10+, Flask 3.x, Modern web browser (Chrome/Firefox/Edge), npm (for optional docx generation), WSL or Linux for development

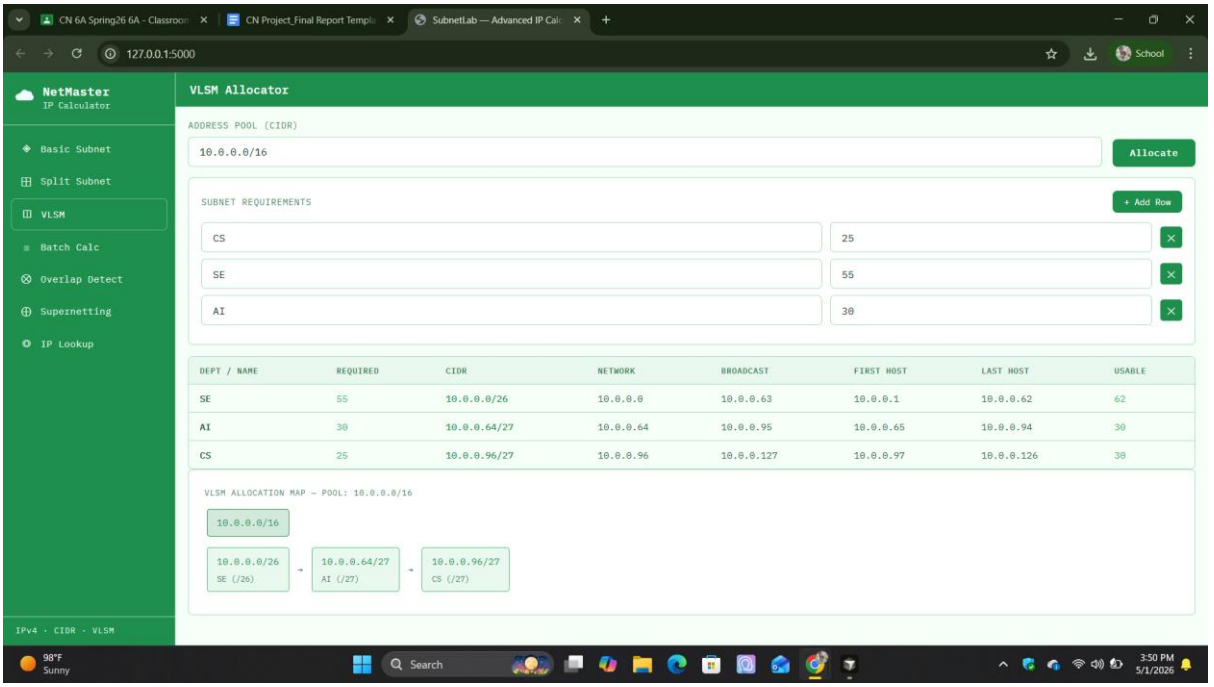
Hardware Requirements

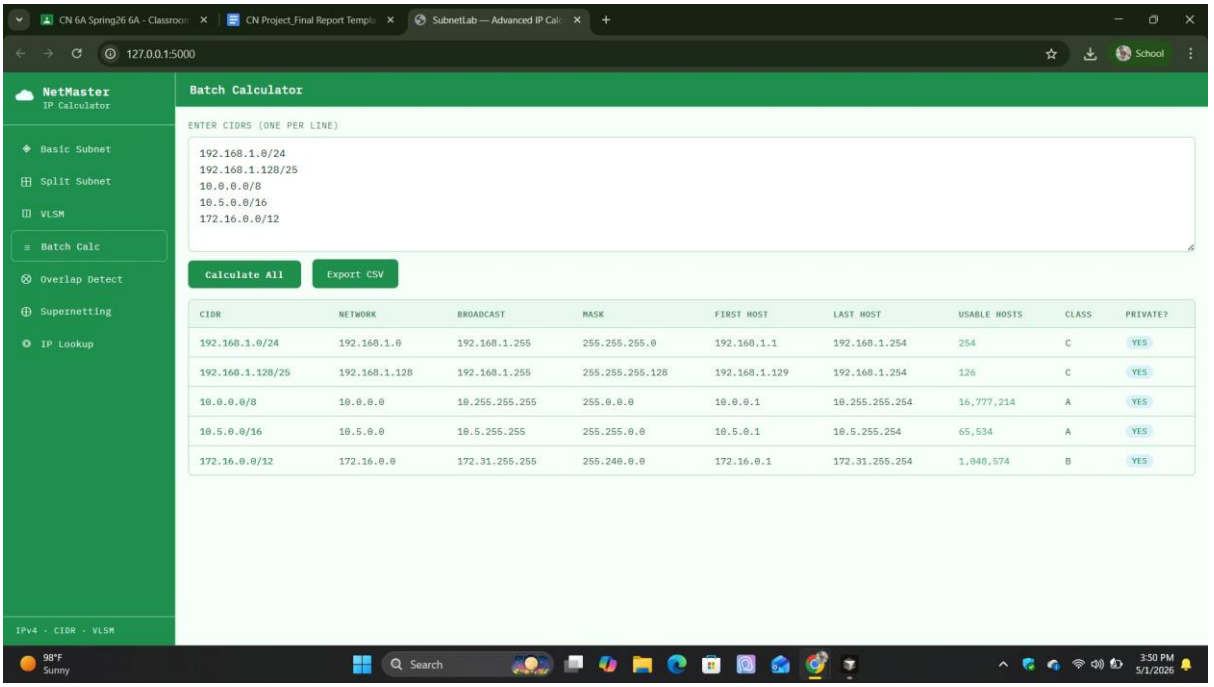
Any standard laptop or PC with at least 4GB RAM. No GPU or specialized hardware required. Network access not required for local deployment

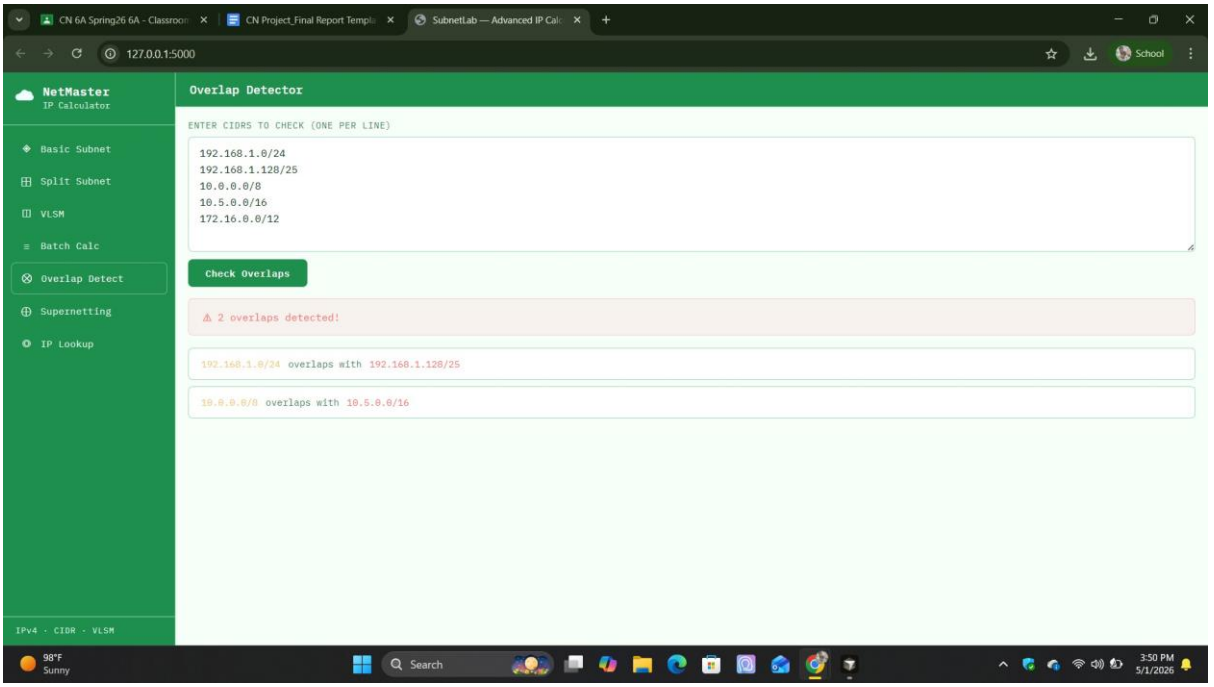
7. Diagrammatic Representation of the Overall System

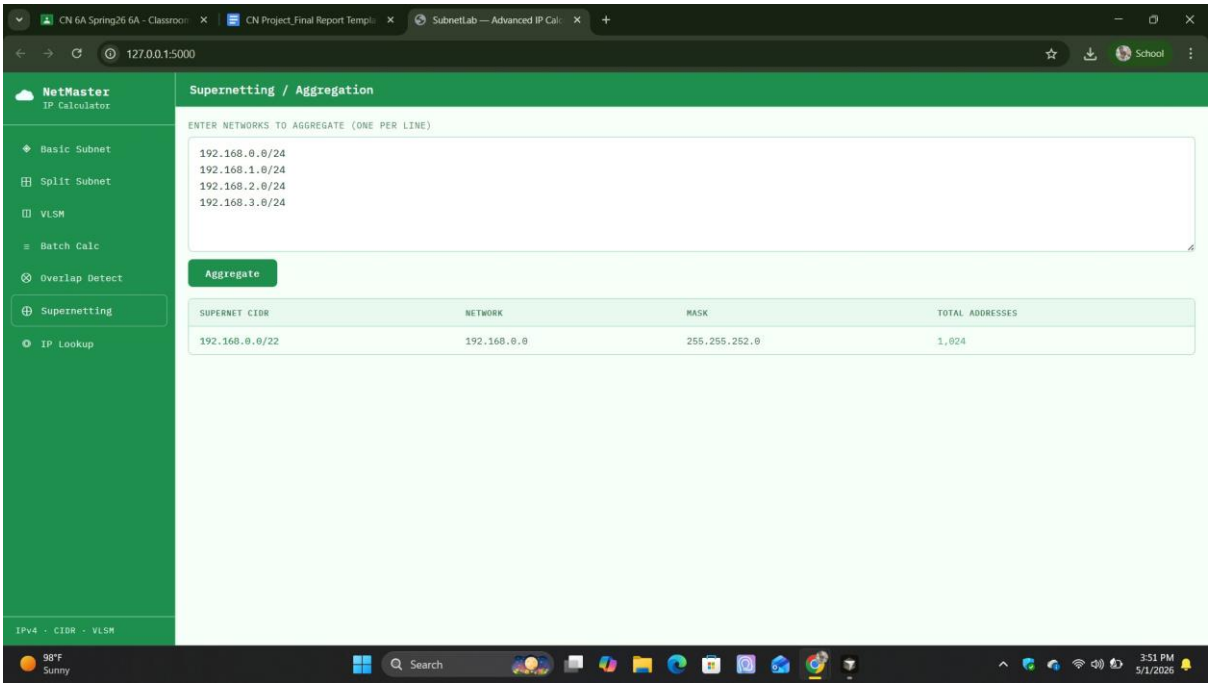


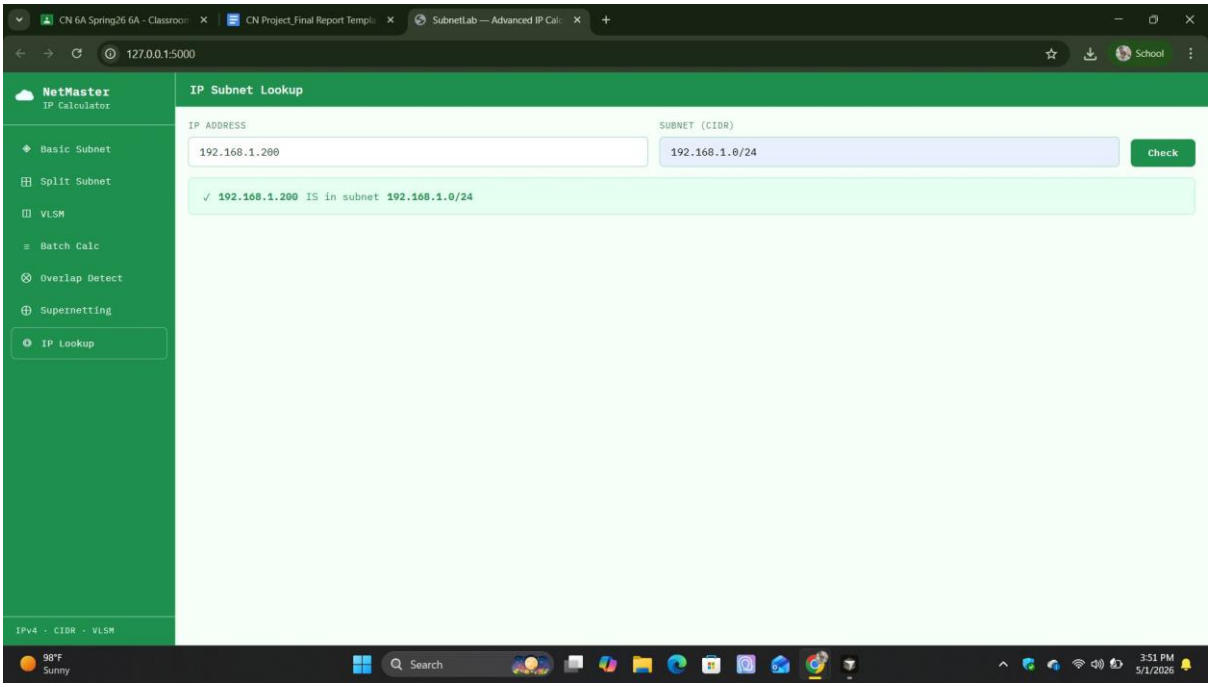

CN 6A Spring26 6A - Classroom
CN Project_Final Report Templa
SubnetLab — Advanced IP Calc
127.0.0.1:5000
School
NetMaster
IP Calculator
Basic Subnet
Split Subnet
VLSM
Batch Calc
Overlap Detect
Supernetting
IP Lookup
IPv4 · CIDR · VLSM
Subnet Splitter
PARENT NETWORK (CIDR)
192.168.10.0/24
SPLIT INTO N SUBNETS
4
Split
SUBNET HIERARCHY
192.168.10.0/24 (parent)
192.168.10.0/26
192.168.10.64/26
192.168.10.128/26
192.168.10.192/26
CIDR NETWORK BROADCAST FIRST HOST LAST HOST USABLE HOSTS
1 192.168.10.0/26 192.168.10.0 192.168.10.63 192.168.10.1 192.168.10.62 62
2 192.168.10.64/26 192.168.10.64 192.168.10.127 192.168.10.65 192.168.10.126 62
3 192.168.10.128/26 192.168.10.128 192.168.10.191 192.168.10.129 192.168.10.190 62
4 192.168.10.192/26 192.168.10.192 192.168.10.255 192.168.10.193 192.168.10.254 62
98°F
Sunny
Search
3:49 PM
5/1/2026


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Overlap Detect
Supernetting
IP Lookup
IPv4 · CIDR · VLSM
VLSM Allocator
ADDRESS POOL (CIDR)
10.0.0.0/16
Allocate
SUBNET REQUIREMENTS
+ Add Row
CS 25
SE 55
AI 30
DEPT / NAME REQUIRED CIDR NETWORK BROADCAST FIRST HOST LAST HOST USABLE
SE 55 10.0.0.0/26 10.0.0.0 10.0.0.63 10.0.0.1 10.0.0.62 62
AI 30 10.0.0.64/27 10.0.0.64 10.0.0.95 10.0.0.65 10.0.0.94 30
CS 25 10.0.0.96/27 10.0.0.96 10.0.0.127 10.0.0.97 10.0.0.126 30
VLSM ALLOCATION MAP — POOL: 10.0.0.0/16
10.0.0.0/16
10.0.0.0/26
SE (/26)
10.0.0.64/27
AI (/27)
10.0.0.96/27
CS (/27)
98°F
Sunny
Search
3:50 PM
5/1/2026


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IP Lookup
IPv4 · CIDR · VLSM
Batch Calculator
ENTER CIDRS (ONE PER LINE)
192.168.1.0/24
192.168.1.128/25
10.0.0.0/8
10.5.0.0/16
172.16.0.0/12
Calculate All
Export CSV
CIDR NETWORK BROADCAST MASK FIRST HOST LAST HOST USABLE HOSTS CLASS PRIVATE?
192.168.1.0/24 192.168.1.0 192.168.1.255 255.255.255.0 192.168.1.1 192.168.1.254 254 C YES
192.168.1.128/25 192.168.1.128 192.168.1.255 255.255.255.128 192.168.1.129 192.168.1.254 126 C YES
10.0.0.0/8 10.0.0.0 10.255.255.255 255.0.0.0 10.0.0.1 10.255.255.254 16,777,214 A YES
10.5.0.0/16 10.5.0.0 10.5.255.255 255.255.0.0 10.5.0.1 10.5.255.254 65,534 A YES
172.16.0.0/12 172.16.0.0 172.31.255.255 255.240.0.0 172.16.0.1 172.31.255.254 1,048,574 B YES
98°F
Sunny
Search
3:50 PM
5/1/2026


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School
NetMaster
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Basic Subnet
Split Subnet
VLSM
Batch Calc
Overlap Detect
Supernetting
IP Lookup
IPv4 · CIDR · VLSM
Overlap Detector
ENTER CIDRS TO CHECK (ONE PER LINE)
192.168.1.0/24
192.168.1.128/25
10.0.0.0/8
10.5.0.0/16
172.16.0.0/12
Check Overlaps
⚠ 2 overlaps detected!
192.168.1.0/24 overlaps with 192.168.1.128/25
10.0.0.0/8 overlaps with 10.5.0.0/16
98°F
Sunny
Search
3:50 PM
5/1/2026


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NetMaster
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Basic Subnet
Split Subnet
VLSM
Batch Calc
Overlap Detect
Supernetting
IP Lookup
IPv4 · CIDR · VLSM
Supernetting / Aggregation
ENTER NETWORKS TO AGGREGATE (ONE PER LINE)
192.168.0.0/24
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24
Aggregate
SUPERNET CIDR
NETWORK
MASK
TOTAL ADDRESSES
192.168.0.0/22
192.168.0.0
255.255.252.0
1,024
98°F
Sunny
Search
3:51 PM
5/1/2026


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NetMaster
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Basic Subnet
Split Subnet
VLSM
Batch Calc
Overlap Detect
Supernetting
IP Lookup
IPv4 · CIDR · VLSM
IP Subnet Lookup
IP ADDRESS
192.168.1.200
SUBNET (CIDR)
192.168.1.0/24
Check
✓ 192.168.1.200 IS in subnet 192.168.1.0/24
98°F
Sunny
Search
3:51 PM
5/1/2026